

AYMANE AIT DADS

Computer Vision & Machine Learning Intern | Deep Learning · Image Classification · PyTorch

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PROFESSIONAL SUMMARY

Ranked 1st in the EURECOM class leaderboard on a 250K-image AI-generated image detection benchmark (0.791 AUC), fine-tuning a 428M-parameter CLIP ViT-L/14 backbone with custom MLP head and 10-view TTA for out-of-distribution generalization. Achieved 99.8% accuracy and ROC AUC of 1.0 on aerial image classification across 17,500+ drone images by benchmarking CNN, DenseNet121, and hybrid CNN-Transformer architectures. Brings academic background in computer vision, deep learning, and signal processing alongside proficiency in Python, PyTorch, and OpenCV-compatible image model building and validation for vision applications. Aims to apply deep learning algorithms and analytical problem-solving to semiconductor wafer defect detection and metrology challenges at Applied Materials.

CORE COMPETENCIES

Computer Vision | Deep Learning | Defect Detection | Image Classification | Algorithm Prototyping | Python Proficiency | ML Model Validation | Signal Processing

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and validated **Random Forest and K-Means ML models** on 100K+ fiber-optic network records to classify network quality across thousands of nodes using Python and Scikit-learn.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built with Pandas and Scikit-learn on structured signal-quality data.
- Delivered stakeholder presentations translating **model outputs into maintenance recommendations**, with visual reporting produced in Power BI for non-technical decision-makers.

PROJECTS

AI-Generated Image Detection - EURECOM

EURECOM Course + International Benchmark · 2025 · Ranked 1st in Class (0.791 AUC)

ImSecu + NTIRE 2026 @ CVPR

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a **CLIP ViT-L/14 backbone** (428M params) with a custom MLP head, dual learning rates, and unfreezing up to 8 transformer blocks for deep learning image classification.
- Built and validated an ensemble of 3 models weighted by validation AUC with **10-view TTA** (flips, crops, blur, resizes) trained on 250K images from 25+ generator types; submitted independently to NTIRE 2026 @ CVPR CodaBench international benchmark.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

Aerial Cactus Detection - Endangered Species Classification

EURECOM · 2025 · 99.8% Accuracy · ROC AUC 1.0

- Prototyped and benchmarked **CNN, DenseNet121, and hybrid CNN-Transformer** architectures on 17,500+ drone images, achieving 99.8% accuracy and F1 = 0.999 on endangered species classification.
- Evaluated deep learning models using ROC AUC, F1-score, and precision-recall metrics, applying **analytical problem-solving** to select optimal architecture for high-stakes aerial image classification.

PyTorch · DenseNet121 · CNN · Scikit-learn · Python

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80 · Unsupervised Fault Detection

- Designed a **Transformer autoencoder on Mel spectrograms** with SpecAugment for unsupervised fault detection on slide-rail industrial machinery with no labeled anomaly examples.
- Achieved **AUC 0.80** for predictive maintenance, applying signal processing theory to transform raw audio waveforms into structured spectral feature representations for deep learning evaluation.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder · Python

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Signal Processing, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Computer Vision & Deep Learning: PyTorch | CLIP ViT | CNN | DenseNet | Image Classification | OpenCLIP | TensorFlow

ML & Algorithmic Prototyping: Python | Scikit-learn | Hugging Face Transformers | Ensemble Methods | Feature Engineering | NumPy | Pandas

MLOps & Data Engineering: FP16/BF16 Mixed Precision | Git | Linux | Kaggle GPU | Power BI | Matplotlib | Seaborn